

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

*Assessment Tool Weightage: 30% Software: Cisco
Packet Tracer, Wireshark*

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly

Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. Debugging IPv4 Network Configuration
Given

- IP Address: **192.168.10.65**
- Subnet Mask: **255.255.255.192 (/26)**
- Default Gateway: **192.168.10.1**

Step 1: Identify the subnet

For /26, block size = 64.

Subnets are:

- 192.168.10.0 – 192.168.10.63
- **192.168.10.64 – 192.168.10.127**
- 192.168.10.128 – 192.168.10.191
- 192.168.10.192 – 192.168.10.255

IP **192.168.10.65** belongs to:

192.168.10.64/26

Step 2: Network Details

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host Range:** 192.168.10.65 – 192.168.10.126

Step 3: Verify IP Address

Configured IP:

192.168.10.65

✓ Valid host address.

Step 4: Verify Gateway

Gateway:

192.168.10.1

This belongs to subnet

192.168.10.0/26

Workstation belongs to

192.168.10.64/26

✗ Gateway is in a different subnet.

Step 5: Configuration Error

- Incorrect default gateway.
- Workstation cannot communicate because the gateway is outside its subnet.

Step 6: Optimized Configuration

Setting	Correct Value
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IP Address	192.168.10.65
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Subnet Mask	255.255.255.192
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Default Gateway	192.168.10.126 (or any valid router IP in 192.168.10.64/26)
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Step 7: Usable Hosts

$$2^{32-26} - 2 = 2^6 - 2 = 64 - 2 = 62$$

Usable Hosts = 62

Final Answer

- Subnet: **192.168.10.64/26**
- Network Address: **192.168.10.64**
- Broadcast Address: **192.168.10.127**
- Host Range: **192.168.10.65 – 192.168.10.126**
- Error: Wrong default gateway
- Correct Gateway: **192.168.10.126**
- Usable Hosts: **62**

2. Debugging IPv6 Configuration

Given

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Step 1: Expand Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

Device A

Network Prefix

2001:db8:0:0::/64

Device B

Network Prefix

2001:db8:0:0::/64

Step 3: Same Subnet?

Yes.

Both have

2001:db8:0:0::/64

✓ Same subnet.

Step 4: Problem

No addressing mismatch.

Communication failure is likely due to:

- IPv6 disabled
- Firewall blocking traffic
- Duplicate address
- Interface down
- Incorrect router configuration

Step 5: Optimized Configuration

Example

Device A

2001:db8:0:0::10/64

Device B

2001:db8:0:0::20/64

Gateway

2001:db8:0:0::1

Step 6: Host Addresses

Host bits = 64

$$2^{64}$$

= 18,446,744,073,709,551,616

Final Answer

- Expanded addresses shown above
- Network Prefix: **2001:db8:0:0::/64**
- Same subnet: **Yes**
- Issue: Not an addressing mismatch; likely another network configuration issue
- Correct example addresses: **2001:db8:0:0::10/64** and **2001:db8:0:0::20/64**
- Available Hosts: **2^{64}**

3. Optimizing Subnet Allocation using VLSM

Given

Network

192.168.1.0/24

Subnets

- /26
- /27
- /28

Current Allocation

/26

Network

192.168.1.0

Broadcast

192.168.1.63

Host Range
192.168.1.1 – 192.168.1.62
Usable Hosts
62
/27
Network
192.168.1.64
Broadcast
192.168.1.95
Host Range
192.168.1.65 – 192.168.1.94
Usable Hosts
30
/28
Network
192.168.1.96
Broadcast
192.168.1.111
Host Range
192.168.1.97 – 192.168.1.110
Usable Hosts
14
Unused Addresses
Remaining
192.168.1.112 – 192.168.1.255

Problem

- Some departments have more IPs than needed.
- Others have too few.
- Address wastage.

Optimized VLSM

Allocate according to requirement:

Department Needed Hosts Recommended Subnet

Largest	50	/26
Medium	25	/27
Small	10	/28

This minimizes unused addresses while meeting current needs.

Final Answer

- /26 → 62 hosts
- /27 → 30 hosts
- /28 → 14 hosts
- Unused addresses should be reassigned using **VLSM** for better utilization.

4. Debugging IPv6 Configuration

This question is identical to **Question 2**.

Answer

- Device A (Expanded): **2001:0db8:0000:0000:0000:ff00:0042:8329**
- Device B (Expanded): **2001:0db8:0000:0000:0001:0000:0000:0001**
- Network Prefix (both): **2001:db8:0:0::/64**
- Same subnet: **Yes**
- No prefix mismatch.
- Suggested addresses:
 - Device A: **2001:db8:0:0::10/64**
 - Device B: **2001:db8:0:0::20/64**
- Available host addresses: $2^{64} = 18,446,744,073,709,551,616$

5. Debugging Routing Table

Routing Table

Destination Network Next Hop

192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3

Destination Network Next Hop

Default Route 10.0.0.1

Destination IP

192.168.2.100

Step 1: Longest Prefix Match

Destination

192.168.2.100

Matches

192.168.2.0/24

Step 2: Next Hop

Selected next hop

10.0.0.3

Step 3: Verify Destination

Subnet

192.168.2.0/24

Host Range

192.168.2.1 – 192.168.2.254

Destination

192.168.2.100

✓ Valid.

Step 4: Possible Routing Errors

If packets are not reaching:

- Wrong next-hop IP
- Next-hop router is down
- Interface is down
- Missing return route
- Incorrect static route

Step 5: Optimization

- Verify next-hop **10.0.0.3** is reachable.
- Enable dynamic routing (OSPF/RIP/EIGRP) if the network is large.
- Remove invalid or duplicate routes.
- Keep routing table updated.

Step 6: Default Route

If no destination matches,

Router uses:

Default Route → 10.0.0.1